

Supplement for RDD results

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Abstract

Quarto document renders the outputs for the full set of RDD results (sensitivity check etc). Results are also saved in /paper-outputs as a .tex fragments so they can be pulled into overleaf directly

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1 Topline

- robust effect for authors a 3rd note (`have_another_note = note_counts[(note_counts >= 3)]`)
- with authors a 2nd note (`have_another_note = note_counts[(note_counts >= 2)]`)
- – robust rd not sig, donut RDD sig for 2 “larger holed donuts”, ordinary OLS is sig with 0.08 threshold,
- RDD is not sharp

2 Setup & Data

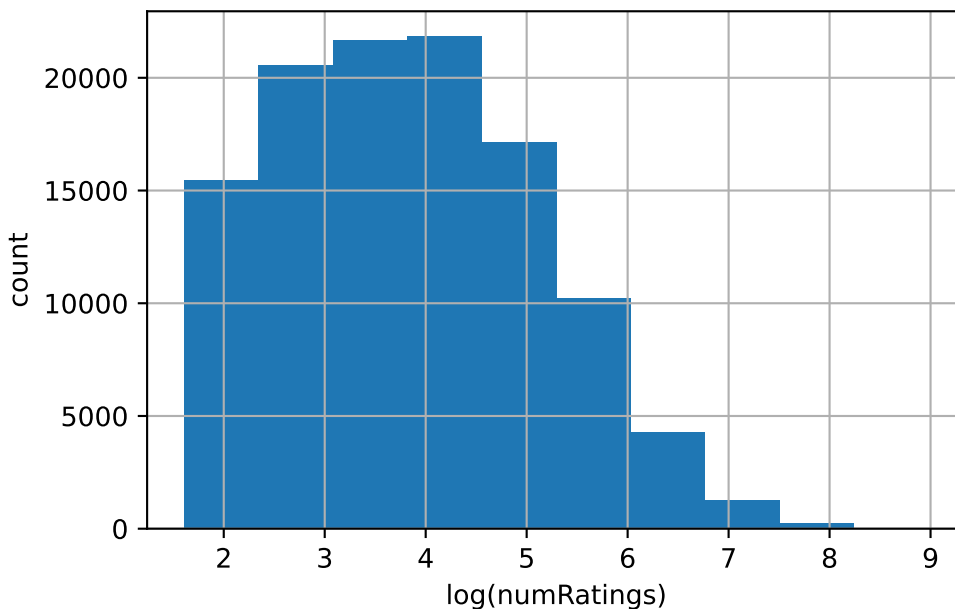
We fix the $\text{cumsum} \geq 2$ issue, so we have authors who write 2 (not 3) notes as our outcome variable
We also remind the alignment filter, since this is conditioning on a collider (and makes our RDD artificially sharp)

```
Imported data for 3173276 notes
ALERT! re_auth_thresh = 2
Now we have 330234 first time note authors (all time)
Now we have 260997 first time note authors 2024-01-01 to 2026-06-02
Now we have 203392 notes written by our 3 choosen algorithms
Now we have 147890 notes classified as MISLEADING
Now we have 112758 notes which have enough ratings
----- No filter for alignment
FINAL total 112758 Notes
of which author writes again = 77111
```

NB current (final), not Max, score

3 Diagnostic Plots

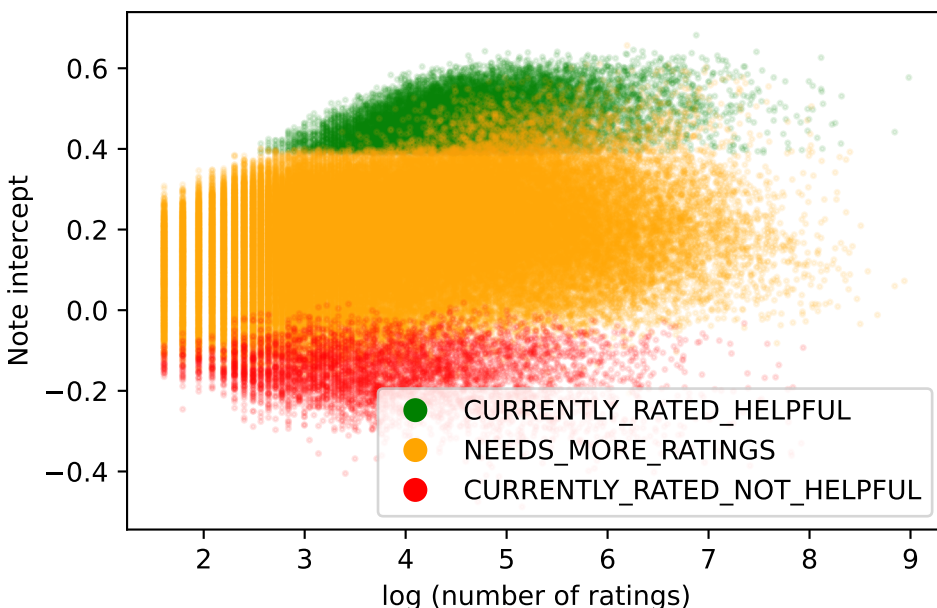
Most notes have a small number of rating, a small number have very many ratings This means the distribution of ratings is very skewed. We take the log of the number of ratings to make the distribution more normal



The basis for our RDD is the discontinuity between the intercept and note status (which defines publication / the possibility for publication)

Plotting $\log(\text{numRatings})$ vs NoteIntercept, with colours according to Note final status (finalRatingStatus)

You can see the fuzzy discontinuity - some >cutoff notes are “NEEDS_MORE_RATINGS”



Sadly, though, our data record intercept calculated from the full set of ratings, this is not the contemporaneous intercept score which determines moment-by-moment publication. It approximates it, but with contamination. Notes may be published then unpublished as they receive more ratings and their rating degrades, or there may be additional blocks on publication (e.g. requirements on total number or type of ratings, or requirements on intercept stability or a lag before publication). The CN system is not entirely clear about all these details.

These contamination can be seen by plotting indicators of ever having been published against our (final) intercept.

We construct an independent ‘ever_published’ variable from the `firstNonNMRStatus` and `mostRecentNonNMRStatus` variables. The logic being that if either of these are `CURRENTLY_RATED_HELPFUL` then the Note must have achieved CRH/published status at some point in the past (ie separate from the `finalRatingStatus` score)

```
[0.35,0.40): n= 4547 P(ever_pub)=0.3851 P(ever_crit)=0.4841
[0.38,0.40): n= 1801 P(ever_pub)=0.5819 P(ever_crit)=0.6680
[0.39,0.40): n= 1145 P(ever_pub)=0.7328 P(ever_crit)=0.7991
[0.40,0.42): n= 2030 P(ever_pub)=0.8788 P(ever_crit)=0.9182
```

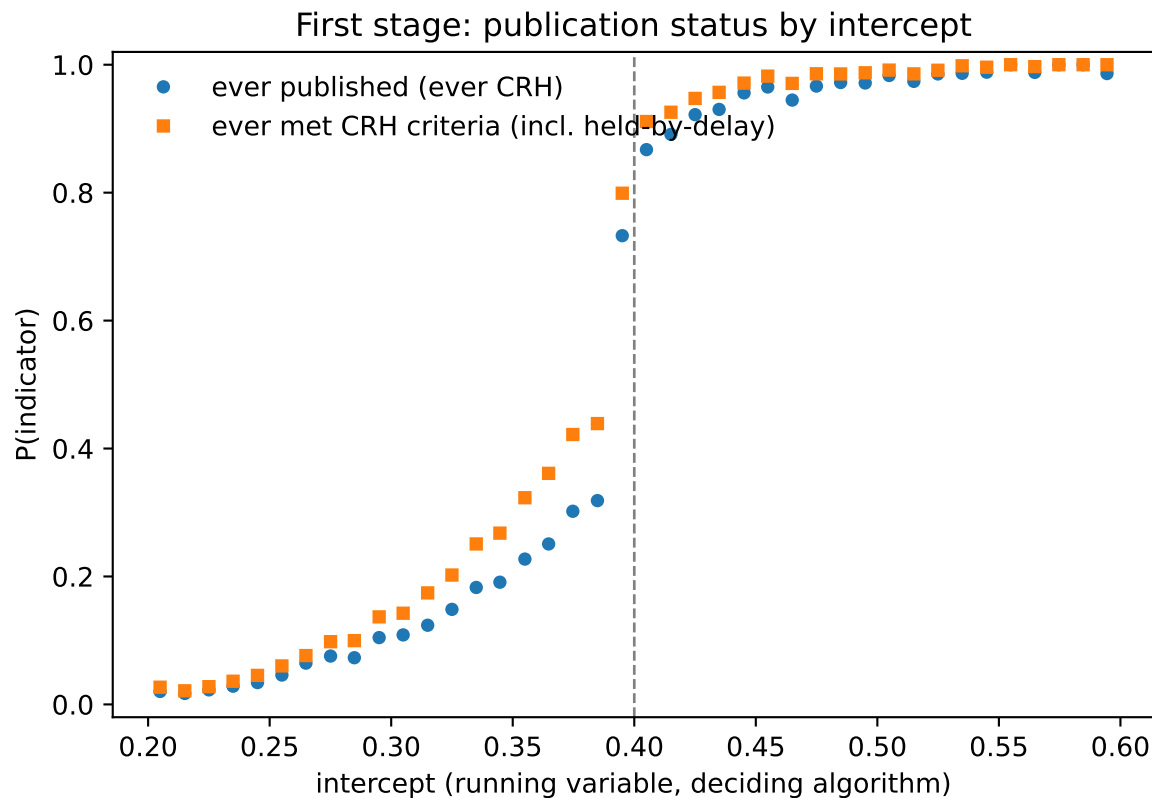
Below-cutoff `P(if_written_again)` by `ever_published`:

	mean	size
<code>ever_published</code>		
0	0.674867	91704
1	0.740338	4321

Below-cutoff `P(if_written_again)` by `ever_met_crh_criteria`:

	mean	size
<code>ever_met_crh_criteria</code>		

False	0.676021	90438
True	0.706819	5587



This shows that our secondary indicator of publication status shows a discontinuity near the cutoff, but not a sharp one

The high proportion of ‘ever published’ just below the cutoff suggests many notes got published and then slipped back below the threshold

We compromise with a fuzzy RDD

4 Trad OLS RDD

Our first approach was to use OLS and an arbitrary window

This was significant with the aligned notes, and a wide window (0.08, it was non-significant at 0.05), but now is not significant at the 0.08 window

bandwidth = 0.08

Number of treated units: 8198

Number of control units: 8829

Estimated Treatment Effect at the Cutoff: 0.0192

I think we can intepret this as a % change the chance of authoring again if published

P-value for treatment: 0.1927
T-statistic for treatment: 1.3027

5 RDD using fuzzy rdrobust with a donut

The rdrobust function will estimate the bandwidth from the data, as well as provide various sensitivity checks and allow a fuzzy RDD

It has the option of using a polynomial fit (which we decline), and uses all points, weighted by proximity to cutoff

We are using a ‘donut’ method of excluding those points near the cutoff, this is specifically to get those points just below the cutoff which have a high (outlier) percentage change of ever having been published

References for this method

@article{calonico2014robust, title = {Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs}, author = {Calonico, Sebastian and Cattaneo, Matias D. and Titiunik, Rocio}, journal = {Econometrica}, volume = {82}, number = {6}, pages = {2295–2326}, year = {2014}, doi = {10.3982/ECTA11757} }

@article{calonico2018effect, title = {On the Effect of Bias Estimation on Coverage Accuracy in Nonparametric Inference}, author = {Calonico, Sebastian and Cattaneo, Matias D. and Farrell, Max H.}, journal = {Journal of the American Statistical Association}, volume = {113}, number = {522}, pages = {767–779}, year = {2018}, doi = {10.1080/01621459.2017.1285776} }

@article{calonico2014rdrobust, title = {Robust Data-Driven Inference in the Regression-Discontinuity Design}, author = {Calonico, Sebastian and Cattaneo, Matias D. and Titiunik, Rocio}, journal = {The Stata Journal}, volume = {14}, number = {4}, pages = {909–946}, year = {2014}, doi = {10.1177/1536867X1401400413} }

Justifying the donut RDD, we can test if intercept (our running variable) is associated with ‘ever published’

PRIMARY SPEC - donut ± 0.01

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations:	108762	
Polynomial Order Est. (p):	1	
Polynomial Order Bias (q):	2	
Kernel:	Triangular	
Bandwidth Selection:	mserd	
Var-Cov Estimator:	NN	
	Left	Right

Number of Observations	94880	13882
Number of Unique Obs.	94731	13864

Number of Effective Obs.	3387	4111
Bandwidth Estimation	0.05	0.05
Bandwidth Bias	0.119	0.119
rho (h/b)	0.418	0.418

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.504	0.032	15.826	2.072e-56	[0.442, 0.567]
Bias-Corrected	0.489	0.032	15.357	3.177e-53	[0.427, 0.552]
Robust	0.489	0.035	13.827	1.740e-43	[0.42, 0.559]

- - - - variation: donut ± 0.0

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations:	110954
Polynomial Order Est. (p):	1
Polynomial Order Bias (q):	2
Kernel:	Triangular
Bandwidth Selection:	mserd
Var-Cov Estimator:	NN

	Left	Right
Number of Observations	96025	14929
Number of Unique Obs.	95875	14909
Number of Effective Obs.	1683	1833
Bandwidth Estimation	0.018	0.018
Bandwidth Bias	0.057	0.057
rho (h/b)	0.318	0.318

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	-0.003	0.027	-0.111	9.113e-01	[-0.057, 0.051]
Bias-Corrected	-0.017	0.027	-0.635	5.255e-01	[-0.071, 0.036]
Robust	-0.017	0.028	-0.615	5.388e-01	[-0.073, 0.038]

- - - - variation: donut ± 0.005

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations:	109838
Polynomial Order Est. (p):	1
Polynomial Order Bias (q):	2
Kernel:	Triangular
Bandwidth Selection:	mserd

Var-Cov Estimator: NN

	Left	Right
Number of Observations	95432	14406
Number of Unique Obs.	95282	14387
Number of Effective Obs.	1184	1477
Bandwidth Estimation	0.02	0.02
Bandwidth Bias	0.07	0.07
rho (h/b)	0.283	0.283

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	-0.123	0.053	-2.307	2.104e-02	[-0.228, -0.019]
Bias-Corrected	-0.153	0.053	-2.851	4.356e-03	[-0.257, -0.048]
Robust	-0.153	0.055	-2.755	5.866e-03	[-0.261, -0.044]

- - - - variation: donut ± 0.02

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 107123
 Polynomial Order Est. (p): 1
 Polynomial Order Bias (q): 2
 Kernel: Triangular
 Bandwidth Selection: mserd
 Var-Cov Estimator: NN

	Left	Right
Number of Observations	94224	12899
Number of Unique Obs.	94075	12883
Number of Effective Obs.	5198	5038
Bandwidth Estimation	0.068	0.068
Bandwidth Bias	0.186	0.186
rho (h/b)	0.368	0.368

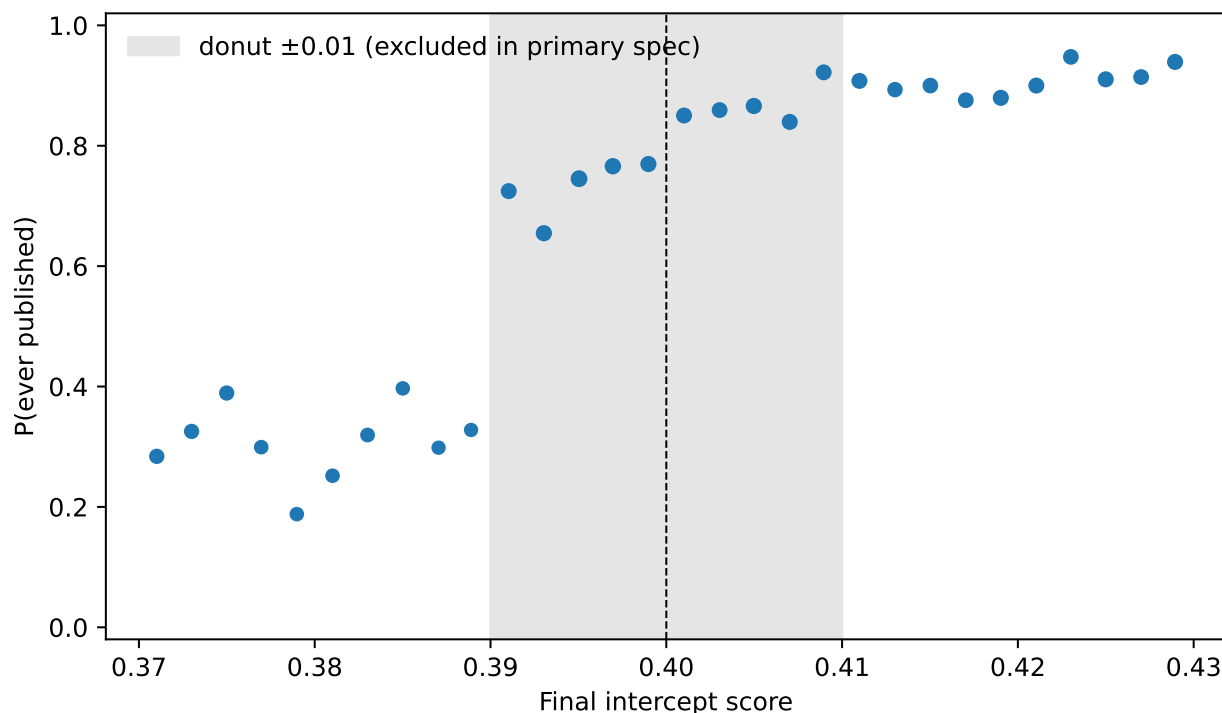
Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.514	0.029	17.439	4.143e-68	[0.456, 0.572]
Bias-Corrected	0.494	0.029	16.738	6.939e-63	[0.436, 0.551]
Robust	0.494	0.032	15.455	6.991e-54	[0.431, 0.556]

So this is saying that if you exclude at 0 (no exclusions) or 0.05 our running variable isn't cleanly associated with "ever published" (at 0.0 it is n.s. at 0.05 it is sig, but negative). At 0.01 the effect is positive and robust (significant), so this justifies the donut

The flip in significance shows significant contamination at the cutoff (those just below *more* likely

to be ever published than those just above!). So the argument is not just excluding problematic points, but the RDD is only possible against a background of linear stability

Note also the negligible drop in N with donut 0.01



'Mephi:\nWhy: the platform has a documented hysteresis rule of exactly 0.01. I just pulled the scoring/src/scoring/mf_base_scorer.py:178,188\nAddCRHInertia rule: a note that is already CRH 1 i.e. [0.39, 0.40). Class in scoring_rules.py:1521, docstring literally says "notes with interce hence the sharp break your plot shows precisely at the grey band\'s left edge.\n'

So this show the operation fo the inertiaDelta = 0.01 rule, see scoring_rules.py line 1521. Notes which are published and drop 0.01 or less keep their status <https://github.com/twitter/communitynotes/blob/>

nb OLS RDD now significant at 0.08 window with cutoff at 0.39, which is nice OR with a cutoff at 0.4 and a donut of 0.01

The OLS serves as a sanity check, but now let's run the full fuzzy RDD

```
running variable (x) length 108762
binary outcome (y) length 108762
cutoff (c) is 0.4
Call: rdrobust
```

Fuzzy RD estimates using local polynomial regression.

```
Number of Observations:      108762
Polynomial Order Est. (p):    1
Polynomial Order Bias (q):    2
Kernel:                      Triangular
```


Bandwidth Selection: mserd
Var-Cov Estimator: NN

	Left	Right
Number of Observations	94880	13882
Number of Unique Obs.	94731	13864
Number of Effective Obs.	6412	6385
Bandwidth Estimation	0.072	0.072
Bandwidth Bias	0.2	0.2
rho (h/b)	0.361	0.361

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.076	0.05	1.543	1.229e-01	[-0.021, 0.174]
Robust	-	-	1.477	1.397e-01	[-0.026, 0.184]

First-Stage Estimates.

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.508	0.021	23.739	1.426e-124	[0.466, 0.55]
Robust	-	-	21.507	1.353e-102	[0.447, 0.537]

MSE-optimal h = 0.0722

Bias-corrected point estimate: 0.0789 95% CI [-0.0258, 0.1836]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0361 (0.5x) tau = +0.1070 95% CI [-0.1141, +0.3204] N_L/N_R = 1945/2684
h = 0.0722 (1.0x) tau = +0.0764 95% CI [-0.0258, +0.1836] N_L/N_R = 6412/6385
h = 0.1444 (2.0x) tau = +0.0970 95% CI [+0.0499, +0.1675] N_L/N_R = 21489/12112

6 Converting rate to baseline outcome rate

Common bandwidth h = 0.0722

Reduced form (ITT): +0.0388 robust CI [-0.0602, +0.1551]
First stage: +0.5082 robust CI [+0.3904, +0.5754]
Fuzzy LATE: +0.0764 robust CI [-0.0258, +0.1836]

Tie-out check: RF / FS = +0.0764 (should LATE at common h)

Mephi suggested text :

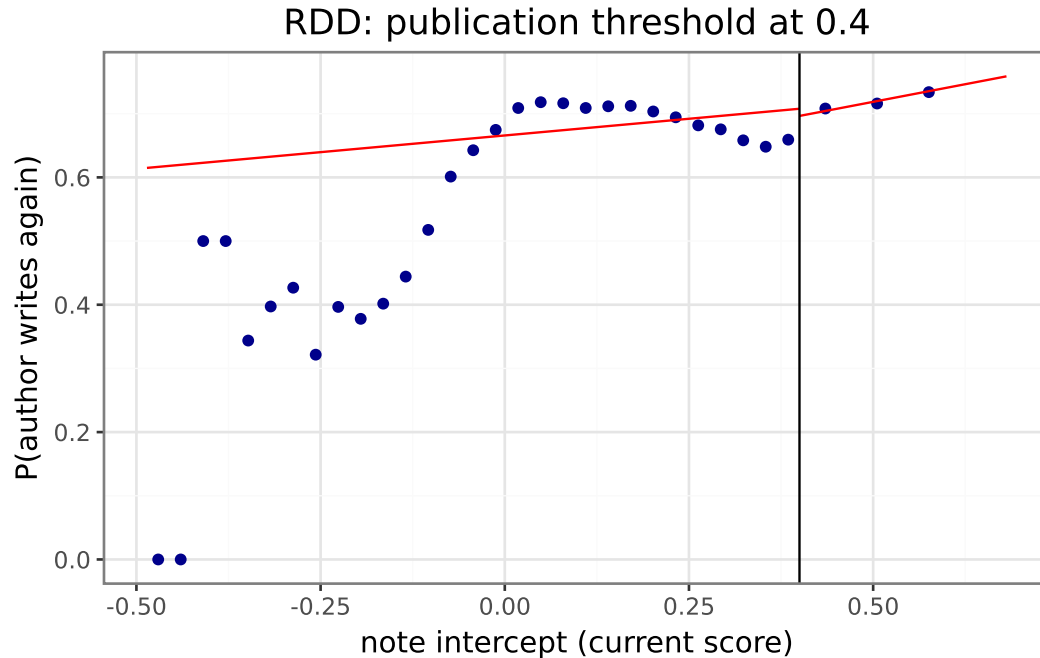
At the MSE-optimal bandwidth (h = 0.072) we estimate $\hat{\tau} = +7.6\text{pp}$ in the probability of writing a second note (conventional local-linear estimator; robust bias-corrected 95% CI [-2.6, +18.4], p = 0.14).

Mephi suggested text: Untreated authors at the cutoff re-author at 65.5%; treated authors at the

cutoff re-author at ~73.2%. Treatment effect = 7.6pp, a 11.7% relative lift over the counterfactual baseline.

7 Showing an illustrative RDD plot

First, binned means with gloabl linear fit (this is the default plot from `rdplot`)



```
[ 'rdplot_mean_bin', 'rdplot_mean_x', 'rdplot_mean_y', 'rdplot_min_bin', 'rdplot_max_bin', 'rdplot_coef' ]
paper-outputs/RDDrobust_scatter.png + paper-outputs/RDDrobust_scatter_data.csv written (32 bins)
```

RD plot at $c = 0.4$ ($p=1$, triangular, $N=108762$)

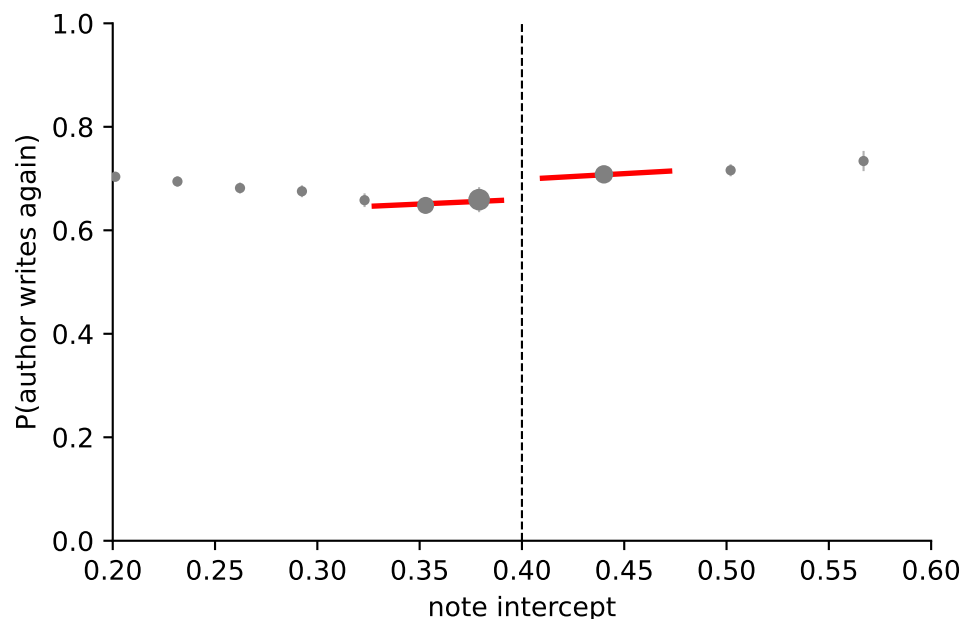
```
bins used   : 29 left / 4 right
J_IMSE      : [29.0, 4.0]
J_MV        : [880.0, 905.0]
scale        : [1, 1]
bin width    : avg [0.03054120705045503, 0.07023575305938719], median [0.030541207050455044, 0.07023575305938719]
```

coefficients:

	Left	Right
0	0.707764	0.696479
1	0.104863	0.221351

wrote `paper-outputs/RDDrobust_scatter_meta.json`, `paper-outputs/RDDrobust_scatter_coef.csv`

Now with the local-linear fits (over the `rdrobust` defined window)



consistency check: gap at cutoff should conventional tau
 plot gap at $c = +0.0388$ vs conventional tau = $+0.0764$

8 Robustness check - density test

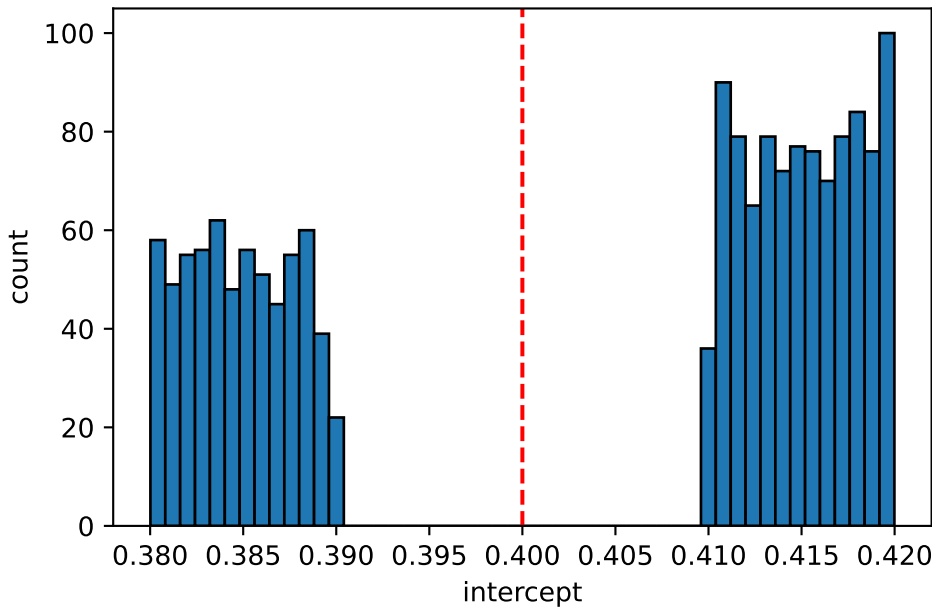
Next, manipulation test/density smoothness at cutoff

if units (people) can select to be just above cutoff then density will not be smooth (think the pass grade on an exam)

Inspecting density with a window of ± 0.02

```
t_asy      NaN
t_jk       0.668638
p_asy      NaN
p_jk       0.503727
dtype: float64
left       0.508816
right      0.627553
diff       0.118737
dtype: float64
left       0.050873
right      0.050371
dtype: float64
full       108762
left       94880
right      13882
eff_left   3485
eff_right  4169
dtype: int64
leftN      [656.0, 883.0, 1168.0, 1460.0, 1735.0, 2042.0, ...]
```

```
rightN      [983.0, 1315.0, 1668.0, 2026.0, 2401.0, 2772.0...
leftWindow   [0.02, 0.023430355260353593, 0.026860710520707...
rightWindow  [0.02, 0.02337457093005241, 0.0267491418601048...
pval         [6.531625492589522e-16, 2.8984895640502804e-20...
dtype: object
['X_max', 'X_min', 'binNW', 'bino', 'binoN', 'binoNStep', 'binoP', 'binoW', 'binoWStep', 'binoWStepN']
```



A wider window shows that the density does progressively decrease up to the cutoff, but no signs of sub-threshold notes being shifted to above threshold (decline not matched by leap post-cutoff)

9 Robustness check - Donut RDD

Exluding results at the cutoff

donut	n_eff_L	n_eff_R	h	coef_conv	se_conv	p_robust	ci_robust_lo	ci_robust_hi
none	6412	6385	0.072	0.0764	0.0495	0.1397	-0.0258	0.1836
±0.005	6412	6385	0.072	0.0764	0.0495	0.1397	-0.0258	0.1836
±0.010	6412	6385	0.072	0.0764	0.0495	0.1397	-0.0258	0.1836
±0.020	5946	5517	0.073	0.0448	0.0647	0.5361	-0.1007	0.1936

10 Sensitivity check - RDD sensitivity with different cutoffs

NB Only a cutoff of 0.4 makes sense in terms of the CN publication mechanism. Therefore we expect other thresholds to show now effect (definitely below 0.4, cutoffs above 0.4 will include some published notes above and below that cutoffs, so if there is a true effect may report an attenuated form of that)

```
running variable (x) length 108762
binary outcome (y) length 108762
cutoff (c) is 0.3
```

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 108762
Polynomial Order Est. (p): 1
Polynomial Order Bias (q): 2
Kernel: Triangular
Bandwidth Selection: mserd
Var-Cov Estimator: NN

	Left	Right
Number of Observations	83588	25174
Number of Unique Obs.	83460	25135
Number of Effective Obs.	13695	8767
Bandwidth Estimation	0.058	0.058
Bandwidth Bias	0.1	0.1
rho (h/b)	0.583	0.583

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.011	0.014	0.81	4.180e-01	[-0.016, 0.038]
Robust	-	-	0.91	3.628e-01	[-0.017, 0.045]

MSE-optimal h = 0.0581

Bias-corrected point estimate: 0.0144 95% CI [-0.0166, 0.0455]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0291 (0.5x) tau = +0.0244 95% CI [-0.0136, +0.0716] N_L/N_R = 6430/5074

h = 0.0581 (1.0x) tau = +0.0110 95% CI [-0.0166, +0.0455] N_L/N_R = 13695/8767

h = 0.1163 (2.0x) tau = +0.0040 95% CI [-0.0121, +0.0332] N_L/N_R = 30357/11892

running variable (x) length 108762

binary outcome (y) length 108762

cutoff (c) is 0.35

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 108762
Polynomial Order Est. (p): 1
Polynomial Order Bias (q): 2
Kernel: Triangular
Bandwidth Selection: mserd
Var-Cov Estimator: NN

Left	Right
------	-------

```

-----
Number of Observations      91478      17284
Number of Unique Obs.      91333      17262
Number of Effective Obs.   22069      8602
Bandwidth Estimation        0.11      0.11
Bandwidth Bias              0.169     0.169
rho (h/b)                   0.652     0.652

```

```

-----
Method      Coef.      S.E.      z-stat      P>|z|      95% CI
-----
Conventional -0.004     0.013    -0.325    7.450e-01   [-0.03, 0.022]
Robust       -         -       -0.486    6.271e-01   [-0.038, 0.023]

```

MSE-optimal h = 0.1100

Bias-corrected point estimate: -0.0075 95% CI [-0.0379, 0.0228]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

```

h = 0.0550 (0.5x) tau = +0.0086 95% CI [-0.0320, +0.0579] N_L/N_R = 8933/3402
h = 0.1100 (1.0x) tau = -0.0043 95% CI [-0.0379, +0.0228] N_L/N_R = 22069/8602
h = 0.2199 (2.0x) tau = +0.0020 95% CI [-0.0225, +0.0249] N_L/N_R = 53603/16546

```

running variable (x) length 13882

binary outcome (y) length 13882

cutoff (c) is 0.45

Call: rdrobust

Sharp RD estimates using local polynomial regression.

```

Number of Observations:      13882
Polynomial Order Est. (p):      1
Polynomial Order Bias (q):      2
Kernel:                      Triangular
Bandwidth Selection:          mserd
Var-Cov Estimator:           NN

```

```

-----
Left      Right
-----
Number of Observations      4140      9742
Number of Unique Obs.      4135      9729
Number of Effective Obs.   1106      1122
Bandwidth Estimation        0.011     0.011
Bandwidth Bias              0.018     0.018
rho (h/b)                   0.577     0.577

```

```

-----
Method      Coef.      S.E.      z-stat      P>|z|      95% CI
-----
Conventional 0.083     0.042     1.968    4.910e-02   [0.0, 0.166]
Robust       -         -         2.016    4.375e-02   [0.003, 0.194]

```

MSE-optimal $h = 0.0106$

Bias-corrected point estimate: 0.0982 95% CI [0.0027, 0.1936]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0053 (0.5x) tau = +0.0737 95% CI [-0.0641, +0.2058] N_L/N_R = 589/579
h = 0.0106 (1.0x) tau = +0.0832 95% CI [+0.0027, +0.1936] N_L/N_R = 1106/1122
h = 0.0212 (2.0x) tau = +0.0410 95% CI [-0.0224, +0.1119] N_L/N_R = 2254/2158

Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 13882
Polynomial Order Est. (p): 1
Polynomial Order Bias (q): 2
Kernel: Triangular
Bandwidth Selection: mserd
Var-Cov Estimator: NN

	Left	Right
Number of Observations	4140	9742
Number of Unique Obs.	4135	9729
Number of Effective Obs.	1106	1122
Bandwidth Estimation	0.011	0.011
Bandwidth Bias	0.018	0.018
rho (h/b)	0.577	0.577

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	0.083	0.042	1.968	4.910e-02	[0.0, 0.166]
Robust	-	-	2.016	4.375e-02	[0.003, 0.194]

All these are non-significant

11 Sensitivity check - data for different time windows

PERIOD 1, from 2025-04-08 to 2026-06-01

ALERT! re_auth_thresh = 2

Now we have 330234 first time note authors (all time)

Now we have 87791 first time note authors 2025-04-08 to 2026-06-01

Now we have 68536 notes written by our 3 choosen algorithms

Now we have 52423 notes classified as MISLEADING

Now we have 34862 notes which have enough ratings

----- No filter for alignment

FINAL total 34862 Notes

of which author writes again = 19514

NB current (final), not Max, score
running variable (x) length 34862
binary outcome (y) length 34862
cutoff (c) is 0.4
Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 34856
Polynomial Order Est. (p): 1
Polynomial Order Bias (q): 2
Kernel: Triangular
Bandwidth Selection: mserd
Var-Cov Estimator: NN

	Left	Right
Number of Observations	29401	5455
Number of Unique Obs.	29391	5448
Number of Effective Obs.	2460	2627
Bandwidth Estimation	0.064	0.064
Bandwidth Bias	0.099	0.099
rho (h/b)	0.642	0.642

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	-0.001	0.03	-0.036	9.712e-01	[-0.06, 0.058]
Robust	-	-	-0.253	8.000e-01	[-0.079, 0.061]

MSE-optimal h = 0.0636

Bias-corrected point estimate: -0.0090 95% CI [-0.0787, 0.0607]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0318 (0.5x) tau = -0.0148 95% CI [-0.1174, +0.0793] N_L/N_R = 1050/1320
h = 0.0636 (1.0x) tau = -0.0011 95% CI [-0.0787, +0.0607] N_L/N_R = 2460/2627
h = 0.1271 (2.0x) tau = +0.0228 95% CI [-0.0381, +0.0637] N_L/N_R = 6786/4626

PERIOD 1, from 2024-01-01 to 2025-04-07

ALERT! re_auth_thresh = 2

Now we have 330234 first time note authors (all time)

Now we have 172844 first time note authors 2024-01-01 to 2025-04-07

Now we have 134572 notes written by our 3 choosen algorithms

Now we have 95257 notes classified as MISLEADING

Now we have 77741 notes which have enough ratings

----- No filter for alignment

FINAL total 77741 Notes

of which author writes again = 57520

NB current (final), not Max, score
running variable (x) length 77741
binary outcome (y) length 77741
cutoff (c) is 0.4
Call: rdrobust

Sharp RD estimates using local polynomial regression.

Number of Observations: 75944
Polynomial Order Est. (p): 1
Polynomial Order Bias (q): 2
Kernel: Triangular
Bandwidth Selection: mserd
Var-Cov Estimator: NN

	Left	Right
Number of Observations	66486	9458
Number of Unique Obs.	66393	9456
Number of Effective Obs.	3370	3667
Bandwidth Estimation	0.059	0.059
Bandwidth Bias	0.103	0.103
rho (h/b)	0.568	0.568

Method	Coef.	S.E.	z-stat	P> z	95% CI
Conventional	-0.006	0.022	-0.262	7.935e-01	[-0.049, 0.038]
Robust	-	-	-0.577	5.641e-01	[-0.063, 0.035]

MSE-optimal h = 0.0586

Bias-corrected point estimate: -0.0144 95% CI [-0.0634, 0.0345]

Bandwidth sensitivity (bias-corrected point, robust 95% CI):

h = 0.0293 (0.5x)	tau = -0.0233	95% CI [-0.0979, +0.0389]	N_L/N_R = 1542/1770
h = 0.0586 (1.0x)	tau = -0.0058	95% CI [-0.0634, +0.0345]	N_L/N_R = 3370/3667
h = 0.1171 (2.0x)	tau = +0.0196	95% CI [-0.0214, +0.0503]	N_L/N_R = 10056/7030